INCH-POUND
MIL-C-L7694/l(SH)
19 October 1990

MILITARY SPECIFICATION SHEET

CABLE AND CABLE ASSEMBLIES, SPECIAL PURPOSE, ELECTRICAL MAGNETIC MINESWEEPING, QUAD CABLE AND QUAD CABLE ASSEMBLIES

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS). specified in the solicitation: MIL-C-17694.

Magnetic minesweeping cables covered by this specification sheet are of the "Quad" type. The quad type cables are designated Q1, Q2 and Q3. Each cable consists of a quad assembly section (Ql, Q2 or Q3) and a single assembly section (S2 or S3). S2 assemblies are used with both Q1 and Q2 minesweeping cables.

The various assembly sections of the quad minesweeping cables have been given the following designations for identification:

TABLE I. Quad minesweeping cable designation.

Quad cable	Non-buoyant quad assembly section	Buoyant quad assembly section	Buoyant single assembly section
Q1	PQ1	AQ1	S2
Q2	PQ2	AQ2	S2
Q3	PQ3	AQ3	S3

Specified standard length requirements for quad cables.

(a) Two of the cables in a PQ quad section shall be longer than the nominal length to permit staggering the PQ-AQ and AQ-AQ pigtail joints as specified by the referenced drawings.
(b) Two of the cables in the PQ3 quad section shall include, as a part of the specified length, 9 foot lengths of 2000 circular roils (MCM) flexible copper pigtails, and the other two shall include 10 foot lengths of 2000 MCM flexible copper pigtails (see type Q3 cable assemblies).

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Quad cables and cable assemblies, general requirements.

(a) For the quad cables (both AQ and PQ), foam polyethylene, fabric reinforced rubber jacketed solid wood cylinders alternated with cellular synthetic robber spacers or extruded rubber should be used for the central core of the quad.
(b) Neoprene cord binders are to be spaced at approximately half quad lay lengths.
(c) The lay of the quad assembly shall be opposite the lay of the outermost layer of conductors.
(d) The end faces of PQ terminal plugs shall be even within 1/4 inch when PQ cable is lying on a flat surface.
(e) Drawings referenced herein shall be in accordance with the latest edition of NAVSEA 0981-LP-052-8190.

TABLE II. PO/AO junction aluminum splice plugs.

Assembly type	Part number	Drawing
Q1	809	NBD-170-1
Q2	810	NBD-169-1
Q3	811	NBD-168-2

TABLE III. Flexibility test reeling drum diameters.

Assembly type	Drum diameter (inches)
Q1	42
Q2	66
Q3	72

CABLE ASSEMBLIES:

Cable assemblies. Each type of cable assemblies (quad assembly and single assembly) shall be in accordance with Drawing S8101-1449611 and the following drawings:

Q1 Cable Assemblies	S8101-1449537 through S8101-1449541, inclusive.
Q2 Cable Assemblies	S8101-1441851 through S8101-1441854, inclusive and S8101-1441856.
Q3 Cable Assemblies	S8101-1449530 through S8101-1449535, inclusive.
S2 Cable Assemblies	S8101-1449537, S8101-1449539, S8101-1449541, S8101-1441851, S8101-1441853, S8101-1441854 and S8101-1441856.
S3 Cable Assemblies	S8101-144953O and S8101-1449533 through S8101-1449535, inclusive.

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PO cable. PQ1, PQ2 and PQ3 cables shall each consist of four insulated nonbuoyant single-conductors cabled together to form a quad.

AQ cable. AQ1, AQ2 and AQ3 cables shall each consist of four insulated, selfbuoyant, single-conductors cabled together to form a quad.

TABLE IV. Drawings for quad cable construction.

Cable type	Drawing
PQ1 AQ1	S8101-1449541
PQ2 AQ2	S8101-1441856
PQ3 AQ3	S8101-1449535

Terminals:

Type Q1 cable assemblies. The two shorter conductors (short legs) of the PQ1 quad section cable shall be terminated at one end and by part 516 as-shown on Drawing S8101-1449538 and table II. The two longer conductors (long legs) of the PQ1 quad section shall be terminated at one end by part 517 as shown on Drawing S8101-1449538 and table II. The other ends of the four conductors shall be continuous with the AQ1 cable as shown on Drawing S8101-1449538 and table II. Each conductor of the other end of the AQ1 cable shall be terminated with a flexible connection (pigtail) consisting of:

(a) Parts 527, 522, and 524, as shown on Drawing S8101-1449539.
(b) Six feet, 6 inches of 400 MCM, 61 by 7/.0306 inch medium hard drawn (MHD) copper cable.
(c) Part 520 as shown on Drawing S8101-1449539.

Type Q2 cable assemblies. The short legs of the PQ2 section shall be terminated at one end by part 556 as shown on Drawing S8101-1441852 and table II. The long legs of the PQ2 section shall be terminated at one end by part 555 as shown on Drawing S8101-1441852 and table 11. The other ends of the four , conductors shall be continuous with the AQ2 cable as shown on Drawing S8101-1441852 and table II. Each conductor of the other end of the AQ2 cable shall be terminated with a pigtail consisting of:

(a) Parts 564, 560, and 561, as shown on Drawing S8101-1441853.
(b) Six feet, 6 inches of 500 MCM, 61 by 7/.0342 inch MHD copper cable.
(c) Part 558 as shown on Drawing S8101-1441853.

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Type Q3 cable assemblies. Two conductors of the PQ3 shall be terminated with a pigtail consisting of:

(a) Parts 531 and 530, as shown on Drawings S8101-1449531 and S8101-1449532.
(b) Nine feet, 0 inch of 2000 MCM, 61 by 19/.0415 inch tin or alloy coated copper.
(c) Part 528 as shown on Drawings S8101-1449531 and S8101-1449532.

The other two conductors of the PQ3 section shall be terminated with a pigtail consisting of:

(a) Parts 531 and 530, as shown on Drawings S8101-1449531 and S8101-1449532.
(b) Ten feet, 0 inch of 2000 MCM, 61 by 19/.0415 inch tin or alloy coated copper.
(c) Part 529 as shown on Drawings S8101-1449531 and S8101-1449532.

These ends of the PQ3 conductors shall be staggered 1 foot as shown on Drawing S8101-1449530. The other ends of the four conductors shall be continuous with the AQ3 cable. Each conductor of the other end of the AQ3 cable shall be terminated with a pigtail consisting of:

(a) Parts 533, 534, and 535m and shown on Drawing S8101-1449533.
(b) Eight feet, 6 inches of 1000 MCM, 61 by 7/.0484 inch MHD copper cable.
(c) Part 536 as shown on Drawing S8101-1449533.

Assembly. The two AQ conductors joined to the two shorter PQ conductors shall form the short leg and the two AQ conductors joined to the two longer PQ conductors shall form the long leg. The short leg shall be connected directly to a K4 electrode. The long leg shall be connected to an S cable which then connects to a K4 electrode. The short and-long legs shall be staggered 1 foot, 3 inches as shown on the following drawings:

TABLE V. Short and long leg staggering.

Type	Applicable drawing
AQ1	S8101-1449537
AQ2	S8101-1441851
AQ3	S8101-1449530

QUAD CABLE SAMPLE SECTIONS:

PQ cable sections. For PQ cable sections, one pair of components shall be separated out, cut to a length of 5 feet, terminated in the same reamer as the existing terminations, and subjected to the tensile strength test. One component of the other pair shall be subjected to the flexing test.

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TABLE VI. PQ and AQ terminal assembly parts.

Terminal	Applicable drawing	Parts
PQ1	S8101-1449538	518 and 517 (Long leg) 518 and 516 (Short leg)
AQ1	S8101-1449539	524 and 522
PQ2	S8101-1441852	557 and 555 (Long leg) 557 and 556 (Short leg)
AQ2	S8101-1441853	561 and 560
PQ3	S8101-1449531	531 and 530
AQ3	S8101-1449533	535 and 534

AQ cable sections. For AQ cable, the pair of conductors forming the short leg shall be separated from the pair of conductors forming the long leg. This short leg pair shall be cut to a length of 5 feet and the cut ends terminated with the same type of flexible connections and terminal fittings as already on the sample, except that the duplex connector shall remain separated. This pair shall then be subjected to the tensile strength test. Three feet shall be cut from the remaining part of one of the short leg components and subjected to the buoyancy test. One of the long leg components shall be subjected to the flexing test. The remaining long leg conductor and its terminal lug shall be subjected to the compression and terminal lug insulation adhesion tests.

Retesting. In the event of failure of any sample to pass the quad tensile test, both the pair of conductors forming the long leg and the pair forming the short leg shall be retested.

Additional examinations. After winding on shipping reels, 25 feet of the PQ cable shall be unreeled and stretched out flat and the PQ terminals checked for evenness. PQ terminals shall be even within 1/4 inch when so examined.

Reels. Separate reusable steel reels for winding type Q cable assemblies (PQ/AQ cables with K4 electrodes attached) shall be manufactured in accordance with the following drawings:

Assembly type	Drawing
Q1	S-8101-1535865
Q2	S-81O1-1535865
Q3	S-8101-1535867

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Packaging. Packaging shall be as specified in MIL-C-17694,

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